

CPO Accelerates, Yet Stock Implications Vary

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What's New?

Following our earlier notes of *CPO/NPO Accelerates on Aug 17*, we believe that Nvidia's CPO scale-out switch progress is faster than expected. On the other hand, for NPO, due to slower than expected AWS T4 rollout, we revise our total OE shipments forecast to 6.5m/42m in 2027E/2028E respectively. We provide our view on recent development and key stocks below.

Comments

Nvidia's Scale-out CPO: Following Nvidia's Aug-14 announcement, Spectrum-X CPO switches have entered mass production. To support the ramp, TSMC has expanded CPO testing/inspection capacity, notably adding Insertion 2/3 equipment. Key components such as FAU are also progressing well, helped by improving multi-plug/unplug yield rates. At the system level, we believe major component suppliers are already on site, enabling cleanroom-level system integration from early Sept. We now forecast Nvidia's CPO switch shipments at 15k/16k/30k/100k for 4Q26/1Q27/2Q27/2027.

Nvidia & AWS Scale-up NPO: Per our *AI Platform Value Shift* report on Aug 6, NVL576-based scale-up CPO/NPO systems are expected to begin adoption in 2H27E, with two switch boards per NVSwitch tray, each board carrying four ASICs, paired with four 3.2T OEs/ASIC. We reiterate our view that CPO (QM5 + 4 OEs) and NPO (QM5 + 4 OEs on sockets) are progressing in parallel, and our view is similar with Street expectations for full NPO adoption at NVL576 launch given NPO maturity. The next to watch is Feynman, which our supply chain research indicate is also being developed in both CPO and NPO variants; packaging-side visibility suggests CPO (QM6 with six 6.4T OEs) remains on track. For AWS, given limited near-term visibility on T4 tape-out, we now shift our Trainium 4 launch to 1Q28. We forecast AWS OE consumption at 0m/12m units for 2H27E/2028E. Incorporating ASIC and AMD volumes, we now expect global scale-up OE demand to reach 6.5m/42m in 2027E/2028E.

Recommendations

- **Likes: 1) Lumentum (LITE Buy)** on CPO acceleration which supports valuation, with CPO/NPO revenue contribution of 12%/26% in CY27/CY28; **2) Browave (3163 TT) & Bizlink (3665 TT Buy)** on CPO's shuffle box of ~8x content vs. NPO's, with production to accelerate in 1Q27E. **3) Semtech (SMTC Buy)** on TIA/Driver for NPO, despite our downward revision, due to tight supply.
- **Dislikes: Largan (3008 TT)** as our analysis indicates EPS contribution of NT\$20/NT\$97 in CY27/CY28, assuming shipments begin in 2H27, suggesting implied 20x CY28 valuation which is not particularly cheap, with further risk on market share given the incumbent supplier's ongoing yield improvements and a growing number of new entrants.

Risks

- 1) AI demand slowdown, 2) supply constraint, 3) competition.

Figure 1: NPO OE Shipments Forecast (mn units)

	2027E	2028E
NPO OE for Scale-up		
Nvidia's 3.2T NPO	6	19.7
AWS's 6.4T NPO	0	12
AMD's 6.4T NPO	0.5	1.7
Other CSPs' NPO		9
Total Scale-up NPO OE	6.5	42.4

Sources: Company data, GF Securities (Hong Kong) Brokerage.

Figure 2: LITE CPO/NPO revenue contribution

	2027E	2028E
CPO Laser Revenue	427	750
NPO Laser Revenue	501	2,550
Total CPO/NPO Rev (USD m)	928	3,300
Revenue Contribution	12%	26%

Sources: Company data, GF Securities (Hong Kong) Brokerage.

Figure 3: SMTC NPO EPS contribution

	2027E	2028E
NPO Market Share	30%	30%
ASP (\$)	80	80
Net Profit (NM 30%) (USD m)	47	305
EPS Contribution (\$)	0.47	3.03

Sources: Company data, GF Securities (Hong Kong) Brokerage.

Figure 4: Largan CPO/NPO EPS contribution

	2027E	2028E
CPO+NPO OE (m units)	10	50
Market share	20%	20%
ASP (FA+PMLA)	100	100
Net Profit (NM 40%) (USD m)	83	403
EPS Contribution (NT\$)	20	97

Sources: Company data, GF Securities (Hong Kong) Brokerage.

Risks

1) AI demand slowdown, 2) supply constraint, 3) competition.

Rating definitions Benchmark: Hang Seng Index (Hong Kong)

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	Hold	Expected stock relative performance ranges between -10% and 10%
	Underperform	Stock expected to underperform benchmark by more than 10%

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